An Introduction to Universal Artificial Intelligence

Chapter 2: Background — Section 2.2
Measure Theory and Probability Theory

Based on Marcus Hutter, Elliot Catt, David Quarel (2024)

Outline

Why probability theory, with minimal measure theory?

This section builds a working understanding of probability theory from four axioms, and derives from them all the familiar properties (the chain rule $\mathrm{P}(A, B) = \mathrm{P}(A|B)\mathrm{P}(B)$, expectations $\mathbb{E}[X]$, independence, and so on) as theorems rather than assumptions. We only reach for the underlying *measure theory* — the rigorous mathematics of assigning "sizes" to sets — when it is unavoidable, most notably when we discuss convergence of random variables later in this chapter.

Throughout, we use $\mathbb{P}$ for probability, $\mathbb{E}$ for expectation, $\mathbb{R}$ for the real numbers, $\mathbb{N}$ for the natural numbers, and $\mathbb{B} = \{0, 1\}$ for bits.

Definition 2.2.1: σ-algebra

Definition 2.2.1 (σ-algebra)

Given a set Ω (whose elements are called *outcomes*), a collection $\mathcal{F}$ of subsets of Ω is called a *σ-algebra* if it satisfies:

1. $\emptyset \in \mathcal{F}$.
2. If $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$. (Closed under countable union)
3. If $A \in \mathcal{F}$, then $\Omega \setminus A =: A^c \in \mathcal{F}$. (Closed under complement)

Plain English. Ω (capital omega) is the set of every possible outcome of an experiment. $\mathcal{F}$ (script F) is a chosen collection of subsets of Ω — the subsets we are willing to call "events" and assign a probability to. The three rules say: the impossible event must be allowed, taking a union of countably many allowed events must again be allowed, and the complement ("everything except") of an allowed event must be allowed. By De Morgan's law $\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c\right)^c$, a σ-algebra is automatically closed under countable intersection too.

Measurable spaces, and the axioms of a probability measure

Definition 2.2.2 (Measurable space)

A *measurable space* is a pair $(\Omega, \mathcal{F})$ where $\mathcal{F}$ is a σ-algebra on Ω.

We call Ω the *sample space*, $\mathcal{F}$ the *event space*, and a set A in the event space $\mathcal{F}$ an *event*.

Axiom 2.2.3 (Probability (semi)measure)

A *probability measure* (or measure) P on a measurable space $(\Omega, \mathcal{F})$ is a function $\mathrm{P} : \mathcal{F} \to [0, 1]$ satisfying:

1. $\mathrm{P}(\Omega) = 1$.
2. For any sequence of pairwise disjoint events $A_1, A_2, \ldots \in \mathcal{F}$ (i.e. $i \neq j \Rightarrow A_i \cap A_j = \emptyset$),

$$\mathrm{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathrm{P}(A_i)$$

Plain English. P assigns to every event a number in $[0, 1]$: the event containing everything (Ω) always has probability 1 ("certainty"), and if you break an event into countably many non-overlapping pieces, the probability of the whole equals the sum of the probabilities of the pieces.

Weakening the axioms: semimeasures

Axiom 2.2.3 (continued): semi-probability and semimeasures

3. If we weaken Axiom 1 to only require $\mathrm{P}(\Omega) \leq 1$, then we call P a *semi-probability measure*, or simply a *semi-probability*.
4. If we weaken the axioms further to only require $\mathrm{P}(\Omega) \leq 1$ and

$$\mathrm{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \geq \sum_{i=1}^{\infty} \mathrm{P}(A_i)$$

   then we call P a *probability semimeasure* or simply a *semimeasure*.

Plain English. A semi-probability only requires the total probability mass to be *at most* 1 (some probability may be "missing", e.g. mass assigned to outcomes that never occur). A semimeasure further only requires the probability of a countable union to be *at least* the sum of the probabilities of the pieces (rather than exactly equal). These weaker notions turn out to be essential later in the book, e.g. when defining the Solomonoff prior over infinite sequences, where the "missing" mass corresponds to programs that never halt.

Probability spaces, and the jargon of set theory

Definition 2.2.4 (Probability Space)

A *probability space* is a triple $(\Omega, \mathcal{F}, \mathrm{P})$ where $(\Omega, \mathcal{F})$ is a measurable space and P is a probability measure on $(\Omega, \mathcal{F})$.

Plain English. A probability measure can be thought of as a function that takes a particular event A in the event space $\mathcal{F}$, and assigns to it a number representing how likely that event is to occur. Given that we express probability through the language of set theory, the table below is a useful dictionary between the two.

Symbol	Set jargon	Probability jargon
Ω	Collection of objects	Sample space
ω	Element in Ω	Singleton event, outcome
A	Subset of Ω	An outcome in A occurs
A^c	Complement of A	No outcome in A occurs
$A \cap B$	Intersection	An outcome in both A and B occurs
$A \cup B$	Union	An outcome in A or B occurs
$A \setminus B$	Set difference	An outcome in A but not B occurs
$A \,\Delta\, B$	Symmetric difference	Outcome in *either* A or B (not both)
$A \subset B$	Inclusion	If an outcome in A, then in B
$\emptyset$	Empty set	Impossible event
Ω	Universal set	Certain event

Discussion: corollaries of the axioms, and σ-algebra examples I

Six observations about Axiom 2.2.3.

1. Ω is always an event, satisfying $\mathrm{P}(\Omega) = 1$.
2. The *impossible event* $\emptyset$ is an event, and $\mathrm{P}(\emptyset) = 0$.
3. Every finite union/intersection of events, and every countable union/intersection of events, is itself an event (via De Morgan's law).
4. The number of axioms is kept to the minimum needed.
5. The axioms do *not* imply every subset of Ω is an event: $\{\Omega, \emptyset\}$ (the *coarse σ-algebra*) is a valid but boring σ-algebra.
6. For finite or countable Ω, we can always choose $\mathcal{F} = 2^{\Omega}$ (every subset is an event). For uncountable Ω this can be subtly inconsistent (see the Lebesgue measure discussion below).

Discussion: corollaries of the axioms, and σ-algebra examples II

Definition 2.2.7 (Generating a σ-algebra)

Given a family of subsets $\mathcal{S} \subseteq 2^{\Omega}$, the *$\sigma$-algebra generated by* $\mathcal{S}$, denoted $\sigma(\mathcal{S})$, is

$$\sigma(\mathcal{S}) \;=\; \bigcap_{\mathcal{C} \text{ is a } \sigma\text{-algebra} \supseteq \mathcal{S}} \mathcal{C}$$

Plain English. $\sigma(\mathcal{S})$ is the *smallest* σ-algebra containing every set in $\mathcal{S}$: we intersect (take the common part of) every σ-algebra that already contains $\mathcal{S}$. If $\mathcal{F}$ is any σ-algebra containing $\mathcal{S}$, then necessarily $\sigma(\mathcal{S}) \subseteq \mathcal{F}$. As an abuse of notation, we write $\sigma(A) := \sigma(\{A\})$ for a single set A.

Example 2.2.8 (The smallest σ-algebra that contains A)

If $A \subseteq \Omega$, then $\sigma(\{A\}) = \{\emptyset, A, A^c, \Omega\}$. For $\Omega = \{1,2,3,4\}$ and $A = \{\{1\},\{2\}\}$,

$$\sigma(A) = \{\emptyset, \{1\}, \{2\}, \{1,2\}, \{3,4\}, \{1,3,4\}, \{2,3,4\}, \Omega\}$$

Discussion: corollaries of the axioms, and σ-algebra examples III

Caratheodory Extension Theorem. Given $(\Omega, \mathcal{F})$ with $\mathcal{F} = \sigma(\mathcal{S})$ for some $\mathcal{S} \subseteq 2^{\Omega}$, we can uniquely identify a measure P solely from the values it takes on each set in $\mathcal{S}$ — we do not need to specify P on every set in $\mathcal{F}$ directly, only on the generating family.

Theorem 2.2.9: derived probability properties I

Theorem 2.2.9 (Probability properties)

Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space, $A, B \in \mathcal{F}$ events, $\{A_n\}_{n=1}^{\infty}$ a collection of events, and $\{D_n\}_{n=1}^{\infty}$ pairwise disjoint events. Then:

$$\begin{aligned}
\mathrm{P}(\emptyset) &= 0 && (1)\\
\mathrm{P}(A \cup B) &= \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B) && (2)\\
\mathrm{P}(A^c) &= 1 - \mathrm{P}(A) && (3)\\
\text{if } A \subseteq B \text{ then } \mathrm{P}(A) &\leq \mathrm{P}(B) && (4)\\
\mathrm{P}(A) &= \mathrm{P}(A \cap B) + \mathrm{P}(A \cap B^c) && (5)\\
\textstyle\sum_{n=1}^{\infty} \mathrm{P}(D_n) &\leq 1 && (6)\\
\mathrm{P}\big(\textstyle\bigcup_{n=1}^{m} A_n\big) &\leq \textstyle\sum_{n=1}^{m} \mathrm{P}(A_n) \quad \text{(Boole's inequality)} && (7)
\end{aligned}$$

Plain English. These are all the "obvious" facts you already believe about probability — e.g. the probability of A or B counts the overlap $A \cap B$ only once, complements have probability 1 minus the

Theorem 2.2.9: derived probability properties II

original, and probabilities of a growing list of events can only decrease their sum bound (Boole's inequality) — and each one is now a proven *consequence* of just the two axioms above, not a new assumption.

Theorem 2.2.9: derived probability properties III

Two more limit properties. If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ increases, or $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ decreases, then

$$\mathrm{P}\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \lim_{n\to\infty} \mathrm{P}(A_n) \qquad \text{or} \qquad \mathrm{P}\Big(\bigcap_{n=1}^{\infty} A_n\Big) = \lim_{n\to\infty} \mathrm{P}(A_n)$$

respectively. In words: the probability of a limit of a monotone sequence of sets equals the limit of the probabilities — the limit can be "pushed through" P.

Proof sketch of Boole's inequality. Let $B_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$ be the "new part" contributed by A_n. Then $\{B_n\}$ are pairwise disjoint, $B_n \subseteq A_n$, and $\bigcup_n A_n = \bigcup_n B_n$. So

$$\mathrm{P}\Big(\bigcup_n A_n\Big) = \mathrm{P}\Big(\bigcup_n B_n\Big) = \sum_n \mathrm{P}(B_n) \leq \sum_n \mathrm{P}(A_n)$$

using Axiom 2.2.3 (countable additivity of disjoint events) for the middle equality, and $\mathrm{P}(B_n) \leq \mathrm{P}(A_n)$ (since $B_n \subseteq A_n$) for the final inequality.

Example 2.2.5: the biased coin

Example 2.2.5 (Biased coin)

To represent flipping a biased coin with bias θ (theta), model with the probability space $(\Omega, \mathcal{F}, \mathrm{P})$, where $\Omega = \{H, T\}$, $\mathcal{F} = 2^{\Omega}$, and

$$\mathrm{P}(\emptyset) = 0, \quad \mathrm{P}(\{H\}) = \theta, \quad \mathrm{P}(\{T\}) = 1 - \theta, \quad \mathrm{P}(\{H, T\}) = 1$$

Plain English. θ is the probability the coin lands heads. Each event is a possible outcome of the coin-flip experiment: $\emptyset$ is "the coin shows neither heads nor tails" (impossible), $\{H\}$ is "heads", $\{T\}$ is "tails", and Ω is "heads or tails" (certain, since the coin must land one way or the other). One can directly verify that $(\Omega, \mathcal{F}, \mathrm{P})$ satisfies Definition 2.2.4.

Example 2.2.6: different choices of Ω for the same experiment

Example 2.2.6 (Different Ω for counting heads)

Flip three coins. *(i)* $\Omega = \{HHH, HHT, HTH, \ldots\}$, $\mathcal{F} = 2^{\Omega}$, distribution $\mathrm{P}(A) = \frac{1}{8}|A|$. The event $E = \{HHT, HTH, THH\}$ (exactly 2 heads) has $\mathrm{P}(E) = \frac{1}{8} \cdot 3 = \frac{3}{8}$. *(ii)* Alternatively, $\Omega' = \{0, 1, 2, 3\}$ (the number of heads shown), $\mathcal{F}' = 2^{\Omega'}$, and

$$\mathrm{P}'(\{0\}) = \tfrac{1}{8}, \quad \mathrm{P}'(\{1\}) = \mathrm{P}'(\{2\}) = \tfrac{3}{8}, \quad \mathrm{P}'(\{3\}) = \tfrac{1}{8}$$

giving the same answer $\mathrm{P}'(\{2\}) = \frac{3}{8}$.

Plain English. Here $|A|$ denotes the number of elements in the set A. Both choices of sample space are equally valid descriptions of "flip three coins"; the second is simply a coarser view that has already grouped outcomes by their number of heads. We construct Ω' by *partitioning* Ω based on the number of heads (e.g. $\{2\} \equiv \{HHT, HTH, THH\}$), and then P' assigns to each group the probability P assigned to the corresponding subset of Ω. This idea — inferring a probability distribution over a partition of Ω — is formalized elegantly by *random variables* (Section 2.2.4).

The Lebesgue measure on $(0, 1)$ I

To model a real number sampled uniformly from the unit interval, choose $\Omega = (0, 1)$ and let $\mathcal{F}$ be the smallest σ-algebra containing every interval $(a, b) \subset \Omega$. We then define $\mathrm{P}((a, b)) := b - a$ (the interval's "length"), and for disjoint intervals $A_1, \ldots, A_n$, define $\mathrm{P}(\cup_i A_i) = \sum_i \mathrm{P}(A_i)$ in line with Axiom 2.2.3. This particular choice of measure is called the *Lebesgue measure*.

It is tempting to try to define the Lebesgue measure for *any* subset E of $(0, 1)$ by covering it with a union of intervals and taking the smallest such covering:

$$\mathrm{P}(E) = \inf \left\{ \textstyle\sum_i (b_i - a_i) : E \subset \cup_i (a_i, b_i) \right\}$$

This is called the *Lebesgue outer measure*. For sets E that satisfy *Carathéodory's criterion* — that for every event S,

$$\mathrm{P}(S) = \mathrm{P}(S \cap E) + \mathrm{P}(S \cap E^c)$$

— we say E is *Lebesgue measurable*, and the set of all such E forms a σ-algebra.

Plain English. There unfortunately exist subsets E of $(0, 1)$ that do *not* satisfy Carathéodory's criterion, for which no consistent notion of "length" $\mathrm{P}(E)$ can be assigned without contradicting the

The Lebesgue measure on $(0, 1)$ II

axioms. This is precisely why we cannot naively take $\mathcal{F} = 2^\Omega$ for uncountable Ω: restricting $\mathcal{F}$ to only the Lebesgue measurable sets sidesteps the problem, at the cost of a little extra bookkeeping.
As a corollary, P assigns the same measure to an interval regardless of whether the endpoints are included:

$$\mathrm{P}([a, b]) = \mathrm{P}((a, b]) = \mathrm{P}([a, b)) = \mathrm{P}((a, b))$$

since each singleton point $\{a\}$ has zero length, so we may freely union in or subtract out finitely many singleton sets without affecting the probability of a set.

(Semi)measures on $\Omega = \mathcal{X}^\infty$

Definition 2.2.11 ((Semi)measures on $\Omega = \mathcal{X}^\infty$)

The most important choice of measurable space $(\Omega, \mathcal{F})$ used throughout this book is the set of infinite sequences $\Omega = \Gamma_\epsilon \equiv \mathcal{X}^\infty$, with event space $\mathcal{F} = \sigma(\{\Gamma_x : x \in \mathcal{X}^*\})$ generated by the *cylinder sets* Γ_x. We denote the (semi)measure as ν (nu) instead of P, abusing notation so that for $x \in \mathcal{X}^*$, $\nu(x) := \nu(\Gamma_x)$. The probability semimeasure Axiom 2.2.3 for $\nu : \mathcal{X}^* \to [0,1]$ restated is

$$\nu(\epsilon) \leq 1 \qquad \text{and} \qquad \nu(x) \geq \sum_{x \in \mathcal{X}} \nu(xa)$$

with equality for measures.

Plain English. $\mathcal{X}$ is a finite alphabet, $\mathcal{X}^*$ the set of all finite strings over it, and $\mathcal{X}^\infty$ the set of *all infinite sequences*. Γ_x (the cylinder set of x) is the event "the sampled infinite sequence starts with the finite string x", and ϵ is the empty string, so $\Gamma_\epsilon = \Omega$. Intuitively, $\nu(x)$ should be read as "the probability that a sampled sequence starts with the string x". Since we may not require exact equality ($\nu(x) = \sum_a \nu(xa)$), some probability mass may be "missing" — this turns out to be exactly what is needed later to model universal priors over programs that may never halt.

Lemma 2.2.12: total mass at each string length

Lemma 2.2.12

For a given n and (semi)measure ν, we have $\sum_{x\in\mathcal{X}^n}\nu(x)\le 1$, with equality for measures.

Plain English. $\mathcal{X}^n$ is the set of all strings of exactly length n. This lemma says: no matter how far we extend our strings, the total probability mass over all strings of that length never exceeds 1 — exactly as we would hope, since these strings partition (or sub-partition) the sample space of infinite sequences.

Proof (by induction on n)

Base case: $\nu(\epsilon)\le 1$ follows immediately from Definition 2.2.11. Inductive step:

$$\sum_{x\in\mathcal{X}^{n+1}}\nu(x)=\sum_{x\in\mathcal{X}^n}\sum_{a\in\mathcal{X}}\nu(xa)\le\sum_{x\in\mathcal{X}^n}\nu(x)\le 1$$

The cylinder sets form a *semi-ring*, so for measures ν we can apply Carathéodory's extension theorem to uniquely obtain a measure $\nu(A)$ for all $\mathcal{F}$-measurable sets A. More advanced semimeasure theory (needed once we go beyond cylinder sets) is deferred to Section 2.8.2 of the book.

Conditional probability

Definition 2.2.13 (Conditional probability)

If A and B are events with $\mathrm{P}(A) > 0$, the *conditional probability* of event B given event A is defined as

$$\mathrm{P}(B|A) := \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)}$$

Plain English. $\mathrm{P}(B|A)$ (read "probability of B given A") answers: *restricting attention only to the outcomes where A happened*, what fraction of those also satisfy B? We divide the probability of both happening by the probability of the conditioning event A alone.

Theorem 2.2.14 (Conditional probability measures are probability measures)

If $(\Omega, \mathcal{F}, \mathrm{P})$ is a probability space and $A \in \mathcal{F}$ with $\mathrm{P}(A) > 0$, then $(\Omega, \mathcal{F}, \mathrm{P}_A)$ is also a probability space, with $\mathrm{P}_A(\cdot) := \mathrm{P}(\cdot|A)$.

Plain English. Conditioning on an event doesn't break any of the axioms — the resulting function P_A is itself a perfectly valid probability measure. Consequently, *any* theorem proved for a general probability measure P automatically also holds for P conditioned on any event A with $\mathrm{P}(A) > 0$.

Random variables: motivation and definition

Usually we do not care about the exact sampled outcome $\omega \in \Omega$, but some derived quantity: the sum of two dice, not the specific pair rolled; a win or loss on a roulette spin, not the specific number. A *random variable* formalizes this: a deterministic function of the outcome.

Definition 2.2.16 (Random variable)

A (real-valued) random variable is a function $X : \Omega \to \mathbb{R}$ that satisfies

$$\{\omega \in \Omega : X(\omega) \le r\} \in \mathcal{F} \qquad \text{for all } r \in \mathbb{R}$$

We call the image $X(\Omega) = \{X(\omega) : \omega \in \Omega\}$ the *alphabet* for X.

Plain English. X is deterministic: given the same ω, X always returns the same real number. There is nothing "random" about X itself — the randomness comes entirely from the fact that ω is sampled according to P. The technical condition ($X \le r$ must always be a legitimate event) ensures that X interacts sensibly with the event space $\mathcal{F}$, so we can ask questions like "what is the probability that $X \le r$?"

The cumulative distribution function

Definition 2.2.17 (Cumulative distribution function (cdf))

The *cumulative distribution function*, (cdf) $F_X : \mathbb{R} \to [0,1]$ for a random variable X is defined as

$$F_X(x) = \mathrm{P}(\{\omega \in \Omega : X(\omega) \leq x\})$$

We may also write $\mathrm{P}(X \leq x)$ or simply $F(x)$.

Plain English. $F_X(x)$ (pronounced "F sub X of x") is the probability that the random variable X takes a value *at most* x. It is always well-defined because Definition 2.2.16 guarantees $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$.

Definition 2.2.18 (Discrete random variable)

X is *discrete* if the alphabet $X(\Omega)$ is countable.

Remark 2.2.15 (Types of random variables)

Random variables may be discrete, continuous, or mixed. We mostly focus on discrete random variables, though the theorems here usually also hold for the continuous case (swapping sums for integrals).

Example 2.2.20: fair coin flips as a random variable

Example 2.2.20 (Fair coin flips)

Take the probability space $(\Omega, 2^\Omega, \mathrm{P})$ with $\Omega = \{HHH, HHT, \dots, TTT\}$ and $\mathrm{P}(A) = |A|/8$. Define the random variable

$$X(\omega) = \text{number of heads in } \omega$$

X is discrete with $X(\Omega) = \{0, 1, 2, 3\}$. For example,

$$p_X(2) := \mathrm{P}(\{\omega \in \Omega : X(\omega) = 2\}) = \mathrm{P}(\{HHT, HTH, THH\}) = 3/8$$

Plain English. We map each of the 8 equally likely coin-flip sequences to the (deterministic) count of heads it contains, then read off the induced probability of each count from the underlying uniform distribution over sequences.

Continuous random variables and probability density functions

Definition 2.2.21 (Continuous random variable)

X is *continuous* if there exists an integrable function $f : \mathbb{R} \to [0, \infty)$ such that $F_X(x) = \int_{-\infty}^{x} f(t)\, dt$.

Definition 2.2.22 (Probability density function (pdf))

For a continuous X, the *probability density function* (pdf) is $p_X(x) := f(x)$ from Definition 2.2.21.

Plain English. "Continuous" here is a slight misnomer: it does not mean X itself is a continuous function, only that its cdf F_X is (in fact *absolutely* continuous, being the integral of an integrable function). Continuous random variables model situations with uncountably many outcomes, e.g. heights of people: asking "what is the probability someone is *exactly* 1.8000... meters tall?" is degenerate (the answer is always zero), so instead we only ask for probabilities over *intervals*, e.g. between 1.79m and 1.81m.

Remark 2.2.24 (Densities may not be bounded by 1)

Unlike the cdf or pmf, a pdf can be unbounded, as long as $\int_{-\infty}^{\infty} p_X(t)\, dt = 1$ still holds. Example: $p_X(x) = -\ln(x)\, [\![0 \leq x \leq 1]\!]$, unbounded as $x \to 0$, yet integrates to 1.

Proposition 2.2.23: relating cdf and pmf

Proposition 2.2.23 (Relating cdf and pdf)

For a discrete random variable X, the cdf F_X and the pmf p_X relate as

$$F_X(x) = \sum_{t \le x} p_X(t)$$

where the sum ranges over all $t \in X(\Omega)$ with $t \le x$.

Plain English. The cdf at x is simply the running total (sum) of all the pmf's probability mass at values up to and including x. Note the parallel with the continuous case, $F_X(x) = \int_{-\infty}^{x} p_X(t)\, dt$ — identical up to swapping a sum for an integral.

Proof

$F_X(x) \overset{(a)}{=} \mathrm{P}\big(\bigcup_{t \in X(\Omega), t \le x} \{\omega : X(\omega) = t\}\big) \overset{(b)}{=} \sum_{t \in X(\Omega), t \le x} \mathrm{P}(X = t) \overset{(c)}{=} \sum_{t \le x} p_X(t)$, where (a) expands the event $X \le x$ into a disjoint union over exact values $t \le x$, (b) uses countable additivity (Axiom 2.2.3), and (c) extends the sum over all $t \le x$ since $p_X(t) = 0$ for $t \notin X(\Omega)$.

Example 2.2.25 (Uniform measure on interval $[0,1]$)

Consider $\Omega = [0,1]$ (a dart thrown at the unit interval). Let $\mathcal{F}$ be the closure of $\{(a,b) : a \le b\}$ under complements and countable unions. Since the dart is equally likely to land anywhere, define $\mathrm{P}((a,b)) = b - a$ and $\mathrm{P}(\{a\}) = 0$ for singletons. For a disjoint union $A = \bigcup_{i=1}^{\infty}(a_i, b_i)$,

$$\mathrm{P}(A) := \sum_{i=1}^{\infty}(b_i - a_i)$$

and complements follow: $\mathrm{P}(A^c) := 1 - \mathrm{P}(A)$.

Plain English. $b - a$ is just the "length" of the open interval (a,b), matching our intuition that the chance a dart lands in a sub-interval is proportional to that sub-interval's length. This is precisely the Lebesgue measure introduced two slides ago, specialized to $[0,1]$.

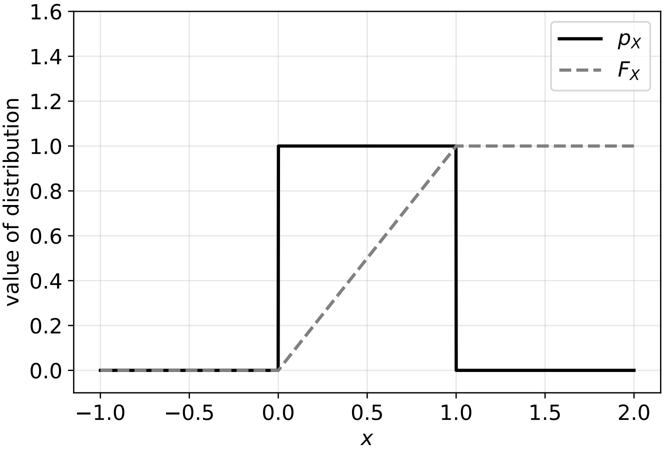
Example 2.2.25: the uniform measure on the unit interval II

Now consider the *identity* random variable $X(\omega) = \omega$ — X simply reports where the dart landed. Its cdf is

$$F_X(x) = \mathrm{P}([0,x]) = \begin{cases} 0 & x < 0 \\ x & 0 \le x < 1 \\ 1 & 1 \le x \end{cases}$$

which gives the pdf $p_X(x) = 1$ for $0 \le x \le 1$, and 0 otherwise. One can verify $\int_{-\infty}^{x} p_X(t)\,dt = F_X(x)$ as required.

Example 2.2.25: the uniform measure on the unit interval III



Example 2.2.25: the uniform measure on the unit interval IV

Plain English (figure). The solid line is the flat pdf (density 1 everywhere on $[0, 1]$, zero outside); the dashed line is the cdf, rising linearly from 0 to 1 across the interval — reflecting that the probability of landing in $[0, x]$ grows in direct proportion to x.

Proposition 2.2.27: specifying a random variable by alphabet and probabilities

Proposition 2.2.27

Let $\{r_i\}_{i=1}^{\infty}$ be distinct real numbers and $\{p_i\}_{i=1}^{\infty}$ a sequence with $p_i \geq 0$ and $\sum_{i=1}^{\infty} p_i = 1$. Then there exists a probability space $(\Omega, \mathcal{F}, \mathrm{P})$ and random variable $X : \Omega \to \mathbb{R}$ such that $X(\Omega) = \{r_n : n \in \mathbb{N}^+\}$ and $\mathrm{P}(X = r_n) = p_n$ for all n.

Plain English. Given *any* desired set of outcomes and matching probabilities, we can always manufacture a probability space and random variable that realizes exactly that distribution. This licenses the common practice of specifying a random variable purely by its alphabet and distribution, without ever mentioning the underlying sample space Ω explicitly — as formalized in the next remark.

Remark 2.2.28 (Discrete random variable via probability mass function)

Equivalently: X is simply a tuple $(\mathcal{X}, p_X)$ with sample space $\mathcal{X}$ and $p_X : \mathcal{X} \to [0, 1]$ satisfying $\sum_{x \in \mathcal{X}} p_X(x) = 1$. We write $X \sim (\mathcal{X}, \mathrm{P}_X)$ or simply $X \sim \mathrm{P}_X$.

Python: random variables, pmf and cdf from scratch

We rebuild Example 2.2.20 (number of heads in 3 fair coin flips) purely from its sample space, and verify Proposition 2.2.23 ($F_X(x) = \sum_{t \le x} p_X(t)$) numerically.

```
import itertools

# Omega: all 8 equally likely 3-coin-flip sequences
omega = list(itertools.product("HT", repeat=3))
P = {w: 1/8 for w in omega}                      # P(A) = |A|/8

def X(w):                                        # random variable: count heads
    return w.count("H")

# Induced pmf p_X(x) = P({w : X(w) = x})
alphabet = [0, 1, 2, 3]
pX = {x: sum(P[w] for w in omega if X(w) == x) for x in alphabet}
print("pmf p_X:", pX)                            # {0: 0.125, 1: 0.375, 2: 0.375, 3: 0.125}

# cdf via Proposition 2.2.23: F_X(x) = sum_{t <= x} p_X(t)
FX = {x: sum(pX[t] for t in alphabet if t <= x) for x in alphabet}
print("cdf  F_X:", FX)                           # {0: 0.125, 1: 0.5, 2: 0.875, 3: 1.0}
```

Output. `pmf p_X: {0: 0.125, 1: 0.375, 2: 0.375, 3: 0.125}`
`cdf  F_X: {0: 0.125, 1: 0.5, 2: 0.875, 3: 1.0}` — matching $\mathrm{P}(X = 2) = 3/8$ from Example 2.2.20 exactly.

Joint cdf, pmf, and pdf

Definition 2.2.29 (Joint cdf, pmf, pdf)

For two random variables X, Y on the same probability space, the *joint cumulative distribution function* (joint cdf) is

$$F_{X,Y}(x, y) := \mathrm{P}(X \leq x \text{ and } Y \leq y)$$

If X, Y are discrete, the *joint probability mass function* (joint pmf) is

$$p_{X,Y}(x, y) := \mathrm{P}(X = x \text{ and } Y = y)$$

Plain English. These extend the cdf/pmf to describe two random variables *simultaneously*: $F_{X,Y}(x, y)$ is the probability that X is at most x *and* Y is at most y, both at once.
For continuous X, Y, assuming suitable regularity, there is an integrable $f : \mathbb{R}^2 \to [0, \infty)$ with $F_{X,Y}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u, v)\, du\, dv$, and we define the *joint pdf* $p_{X,Y}(x, y) := f(x, y)$.

Conditional cdf, pmf, and pdf

Definition 2.2.30 (Conditional cdf, pmf, pdf)

$$F_{Y|X}(y|x) = \mathrm{P}(Y \leq y \mid X = x), \qquad \mathrm{P}(Y = y|X = x) = \frac{\mathrm{P}(Y = y, X = x)}{\mathrm{P}(X = x)}$$

Plain English. These describe the distribution of Y once we already know the exact value taken by X.

Theorem 2.2.31: the sum rule and product rule

Theorem 2.2.31 (Sum and product rule)

For discrete X, Y with outcomes $x \in X(\Omega)$, $y \in Y(\Omega)$:

$$\text{Sum rule:} \quad \mathrm{P}(X = x) = \sum_{y \in Y(\Omega)} \mathrm{P}(X = x, Y = y)$$

$$\text{Product rule:} \quad \mathrm{P}(X = x, Y = y) = \mathrm{P}(X = x | Y = y)\, \mathrm{P}(Y = y)$$

For continuous X, Y with pdfs p_X, p_Y and joint pdf $p_{X,Y}$:

$$\text{Sum rule:} \quad p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x, y)\, dy \qquad \text{Product rule:} \quad p_{X,Y}(x, y) = p_{X|Y}(x|y)\, p_Y(y)$$

Plain English. The *sum rule* says: to get the marginal (i.e. un-conditioned) distribution of X alone, add up (or integrate out) the joint distribution over every possible value of Y. The *product rule* is just Definition 2.2.13 rearranged: the joint probability of $X = x$ and $Y = y$ equals the conditional probability of X given Y, times the (marginal) probability of Y. These also hold conditioned on a third random variable $Z = z$, by Theorem 2.2.14.

Example 2.2.32: conditional coin flips

Example 2.2.32 (Conditional coin flips)

Recall X = number of heads in 3 coin flips. Define $Y(\omega) = [\![\omega \neq TTT]\!]$ ("at least one head was flipped"). We ask: how likely is it that two heads were flipped, given that at least one was?

$$\mathrm{P}(X = 2 | Y = 1) = \frac{\mathrm{P}(X = 2, Y = 1)}{\mathrm{P}(Y = 1)} = \frac{3/8}{7/8} = \frac{3}{7}$$

Plain English. $[\![\cdot]\!]$ is the Iverson bracket: it equals 1 if the statement inside is true, 0 otherwise. The intuition behind $3/7$: knowing $Y = 1$ rules out only the single outcome TTT, leaving 7 equally likely outcomes $\{HHH, HHT, HTH, HTT, THH, THT, TTH\}$, of which exactly 3 have 2 heads.

Example 2.2.34 (Dependent random variables)

X and Y above are *not* independent, since $\mathrm{P}(X = 0, Y = 1) = \mathrm{P}(\emptyset) = 0$ but $\mathrm{P}(X = 0)\mathrm{P}(Y = 1) = \frac{1}{8} \cdot \frac{7}{8} = \frac{7}{64} \neq 0$.

Definition 2.2.33: independent and identically distributed

Definition 2.2.33 (Independent and identically distributed)

A countable collection $(X_i)_{i \in I}$ with cdfs $\{F_{X_i}\}_{i \in I}$ are *identically distributed* if all cdfs are identical. The family (X_n) is *independently distributed* (or *mutually independent*) if for all finite subsets $J \subseteq I$ and all $\{x_j\}_{j \in J}$,

$$F_{X_j : j \in J}(X_j = x_j : j \in J) = \prod_{j \in J} F_{X_j}(X_j = x_j)$$

This is stronger than *pairwise independence*, which only requires $F_{X_i, X_j}(x_i, x_j) = F_{X_i}(x_i) F_{X_j}(x_j)$ for all $i, j \in I$. If both conditions hold, we say the family is *independent and identically distributed* (i.i.d.).

Plain English. "Identically distributed" means every variable in the family follows the exact same probability law. "Independent" means knowing the values of any finite subset of the variables tells you nothing about the others — their *joint* cdf factorizes into the product of individual cdfs. Mutual independence (all finite subsets factorize) is a strictly stronger requirement than mere pairwise independence (only pairs factorize).

Motivating expectation via the empirical average

Given a random variable X, sample outcomes $\omega_1, \omega_2, \ldots$ and feed them into X to get $X(\omega_1), X(\omega_2), \ldots$, then define the *empirical average* $\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X(\omega_i)$. For large n, if $\mathrm{P}(X = x_i) = p_i$ then approximately np_i of the outcomes equal x_i, so

$$\bar{X}_n \;\approx\; \frac{1}{n}\sum_{x\in X(\Omega)} (nP(X=x))\,x \;=\; \sum_{x\in X(\Omega)} x\,P(X=x)$$

Plain English. This heuristic argument (make it rigorous later via the law of large numbers) motivates defining the "theoretical average" value of X as a weighted sum of outcomes, each weighted by its probability of occurring — which is exactly the definition of expectation below.

Definition 2.2.35: expected value

Definition 2.2.35 (Expected value)

The *expected value* (expectation, or mean) of a discrete random variable X with pmf $p(x)$ is

$$\mathbb{E}[X] := \sum_{x \in X(\Omega)} x\, p(x)$$

For continuous X with pdf $p(x)$: $\mathbb{E}[X] := \int_{-\infty}^{\infty} x\, p(x)\, dx$.

Plain English. $\mathbb{E}[X]$ ("expectation of X") is the probability-weighted average of all possible values X can take. Often we write $\sum_{x \in \mathbb{R}}$ or $\sum_x$ instead of $\sum_{x \in X(\Omega)}$, understanding that $p(x) = 0$ outside $X(\Omega)$. This sum/integral may fail to converge, in which case $\mathbb{E}[X]$ is *undefined* — e.g. the Cauchy distribution $p(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2}$ has no defined expected value.

Conditional and multi-variable expectation

Definition 2.2.36 (Conditional expectation)

Given X, Y and $g : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}[g(X)|Y = y] = \sum_{x \in X(\Omega)} g(x)\, p_{X|Y}(x|y) \qquad \text{(discrete)}$$

$$\mathbb{E}[g(X)|Y = y] = \int_{-\infty}^{\infty} g(x)\, p_{X|Y}(x|y)\, dy \qquad \text{(continuous)}$$

Plain English. This weights the possible values of a *function* of X (rather than X itself) by the probability of X's value *conditioned* on $Y = y$. Setting g to the identity and Y to a constant recovers the unconditional $\mathbb{E}[X]$ from Definition 2.2.35.

Definition 2.2.37 (Multi-variable expectation)

For $X_1, \ldots, X_n$ with joint pmf/pdf $p_{X_1,\ldots,X_n}$ and $f : \mathbb{R}^n \to \mathbb{R}$,

$$\mathbb{E}[f(X_1, \ldots, X_n)] := \sum_{\mathbf{x} \in \prod_{i=1}^{n} X_i(\Omega_i)} f(\mathbf{x})\, p_{X_1,\ldots,X_n}(\mathbf{x})$$

Example 2.2.38 and Theorem 2.2.39: linearity of expectation

Example 2.2.38 (Expected value of a fair die)

For X with alphabet $\{1, \ldots, 6\}$, $\mathrm{P}(x) = \frac{1}{6}$,

$$\mathbb{E}[X] = \sum_{x \in \{1,\ldots,6\}} \frac{x}{6} = \frac{7}{2}$$

Theorem 2.2.39 (Expectation is linear)

For random variables X, Y, Z and scalars $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + bY \mid Z = z] = a\,\mathbb{E}[X|Z = z] + b\,\mathbb{E}[Y|Z = z]$$

Plain English. The expectation of a scaled sum is the scaled sum of the expectations — true *regardless* of whether X and Y are independent. This is proven directly from the definition, by factoring out the constants a, b and applying the sum rule (Theorem 2.2.31) to split the double sum over x, y into two separate sums.

Theorem 2.2.40: expectation of independent random variables

Theorem 2.2.40 (Expectation of independent random variables)

If X and Y are independent, $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$.

Plain English. For *independent* variables (and only then, in general) the expectation of the product equals the product of the expectations. The converse is *not* true: $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ does not imply X, Y are independent.

Proof

$$\mathbb{E}[XY] = \sum_{x,y} xy\,\mathrm{P}(X = x, Y = y) = \sum_{x,y} xy\,\mathrm{P}(X = x)\mathrm{P}(Y = y) = \Big(\sum_{x} x\,\mathrm{P}(X = x)\Big)\Big(\sum_{y} y\,\mathrm{P}(Y = y)\Big)$$

using independence ($\mathrm{P}(X = x, Y = y) = \mathrm{P}(X = x)\mathrm{P}(Y = y)$) in the second step, then splitting the double sum into a product of two single sums.

Definition 2.2.41 and Example 2.2.42: variance

Definition 2.2.41 (Variance)

$\mathrm{Var}[X] := \mathbb{E}\big[(X - \mathbb{E}[X])^2\big]$

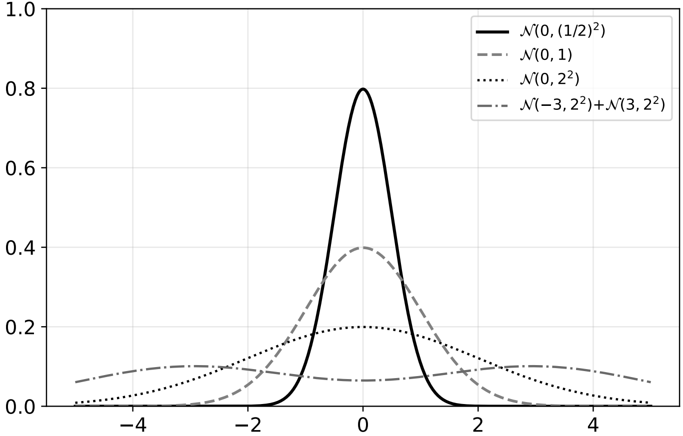
Plain English. Variance measures how far, on average, X strays from its own mean $\mathbb{E}[X]$, using the *squared* error $(X - \mathbb{E}[X])^2$ as the measure of distance (squaring makes positive and negative deviations both count, and penalizes large deviations more). Low-variance random variables concentrate their probability mass near the mean; high-variance ones spread it further away (see the figure on the next slide).

Example 2.2.42 (Expectation and variance of a biased coin)

For $X \in \{0, 1\}$ with $\mathrm{P}(X = 1) = p$, $\mathrm{P}(X = 0) = 1 - p$:

$$\mathbb{E}[X] = p \cdot 1 + (1 - p) \cdot 0 = p \qquad \mathrm{Var}[X] = (1 - p)(0 - p)^2 + p(1 - p)^2 = p(1 - p)$$

Same mean, increasing variance



Plain English (figure). Four densities, all sharing mean 0 but with increasing spread (listed in order of increasing variance): a narrow Gaussian, a wider Gaussian, an even wider Gaussian, and a two-humped mixture of Gaussians centered at ± 3 — the more spread out the mass, the higher the variance.

Theorem 2.2.43: properties of variance I

Theorem 2.2.43 (Properties of variance)

Let X be a random variable, and $X_1, \ldots, X_n$ pairwise independent. Then for any $a \in \mathbb{R}$:

$$\mathrm{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\mathrm{Var}[aX] = a^2 \,\mathrm{Var}[X]$$

$$\mathrm{Var}[X_1 + \cdots + X_n] = \mathrm{Var}[X_1] + \cdots + \mathrm{Var}[X_n]$$

Plain English. The first identity is a convenient computational shortcut ("mean of the square minus square of the mean"). The second says scaling a random variable by a scales its variance by a^2 (variance has units of "squared" X). The third — crucially only valid for *(pairwise) independent* variables — says variances of independent variables simply add when the variables are summed.

Theorem 2.2.43: properties of variance II

Proof. *(i)* Let $b = \mathbb{E}[X]$ (a constant). By linearity,

$$\mathrm{Var}[X] = \mathbb{E}[(X-b)^2] = \mathbb{E}[X^2 - 2bX + b^2] = \mathbb{E}[X^2] - 2b^2 + b^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

(ii) Apply (i) to aX: $\mathrm{Var}[aX] = \mathbb{E}[a^2X^2] - (a\mathbb{E}[X])^2 = a^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) = a^2\mathrm{Var}[X]$. *(iii)* Using $(\sum_i X_i)^2 = \sum_{i,j} X_iX_j$ and (i) for $X := \sum_i X_i$:

$$\mathrm{Var}\Big[\sum_i X_i\Big] = \sum_{i,j} \mathbb{E}[X_iX_j] - \Big(\sum_i \mathbb{E}[X_i]\Big)^2 = \sum_{i,j} \big(\mathbb{E}[X_iX_j] - \mathbb{E}[X_i]\mathbb{E}[X_j]\big)$$

For $i \neq j$, Theorem 2.2.40 (independence) gives $\mathbb{E}[X_iX_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] = 0$, so only the $i = j$ diagonal terms survive:

$$\mathrm{Var}\Big[\sum_i X_i\Big] = \sum_i \big(\mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2\big) = \sum_i \mathrm{Var}[X_i]$$

Theorem 2.2.43: properties of variance III

Remark 2.2.53

Since x^2 is convex, Jensen's inequality (below) gives $\mathbb{E}[X]^2 \leq \mathbb{E}[X^2]$, so $\mathrm{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ is always non-negative.

Predicates, and Markov's inequality

Definition 2.2.44 and Lemma 2.2.45 (Probability of predicates)

For a predicate $S : \Omega \to \{\text{False}, \text{True}\}$, identify False $= 0$, True $= 1$, and define $\mathrm{P}(S) := \mathrm{P}(\{\omega \in \Omega : S(\omega)\})$. Then the random variable $X(\omega) = [\![S(\omega)]\!]$ satisfies $\mathbb{E}[X] = \mathrm{P}(S)$.

Plain English. A predicate is just a True/False statement about the outcome; treating True/False as $1/0$, the expected value of the indicator of S is exactly the probability that S holds.

Theorem 2.2.46 (Markov's inequality)

If a non-negative random variable X has expectation $\mathbb{E}[X]$, then for any $\varepsilon > 0$ (epsilon),

$$\mathrm{P}(X \geq \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon}$$

Plain English. Markov's inequality bounds how much probability mass can sit far out in the tail of a *non-negative* random variable, using only its mean: the chance that X is at least ε can never exceed $\mathbb{E}[X]/\varepsilon$.

Proof of Markov's inequality

Proof of Theorem 2.2.46

Let $Y = \varepsilon[\![X \geq \varepsilon]\!]$. Then $Y \leq X$ always, so $\mathbb{E}[Y] \leq \mathbb{E}[X]$. But $\mathbb{E}[Y] = \varepsilon\,\mathrm{P}(X \geq \varepsilon)$ by Lemma 2.2.45, giving $\mathrm{P}(X \geq \varepsilon) = \mathbb{E}[Y]/\varepsilon \leq \mathbb{E}[X]/\varepsilon$.

Plain English. We build an auxiliary random variable Y that is ε whenever X is at least ε, and 0 otherwise; Y never exceeds X, so its expectation cannot exceed $\mathbb{E}[X]$ either, and $\mathbb{E}[Y]$ is exactly ε times the probability we want to bound.

Chebyshev's inequality

Theorem 2.2.47 (Chebyshev's inequality)

Let X have finite $\mathbb{E}[X]$ and $\mathrm{Var}[X]$. Then for any $\varepsilon > 0$,

$$\mathrm{P}(|X - \mathbb{E}[X]| \geq \varepsilon) \leq \frac{\mathrm{Var}[X]}{\varepsilon^2}$$

Plain English. Chebyshev's inequality bounds how likely X is to land far from its *own* mean, using only its variance. Unlike Markov's inequality, it needs no non-negativity assumption and bounds *both* tails of the distribution at once.

Proof

Let $Y = (X - \mathbb{E}[X])^2 \geq 0$, so $\mathbb{E}[Y] = \mathrm{Var}[X]$. Apply Markov's inequality to $\mathrm{P}(Y \geq \varepsilon^2)$:

$$\mathrm{P}(|X - \mathbb{E}[X]| \geq \varepsilon) = \mathrm{P}((X - \mathbb{E}[X])^2 \geq \varepsilon^2) = \mathrm{P}(Y \geq \varepsilon^2) \leq \frac{\mathbb{E}[Y]}{\varepsilon^2} = \frac{\mathrm{Var}[X]}{\varepsilon^2}$$

Comparing the two. Markov's inequality only bounds the upper tail and needs non-negativity; its bound shrinks inversely in ε. Chebyshev's bounds both tails, needs no non-negativity, and its bound shrinks inversely in ε^2 (i.e. faster).

Python: Markov and Chebyshev bounds vs. exact probability

Example 2.2.48 (Sum of n dice)

$S = \sum_{i=1}^{n} X_i$, $\mathbb{E}[S] = 3.5n$. Markov: $\mathrm{P}(S \geq k) \leq \mathbb{E}[S]/k$. For $n = 20, k = 80$: bound $= 70/80 = 0.875$, true value ≈ 0.1075.

```
import math
n, k = 20, 80
mu = 3.5 * n                    # E[S] for n fair dice
var = n * 35 / 12               # Var[S]; Var of one die = 35/12
# Exact P(S >= k): convolve n copies of a die's pmf
dist = [1.0]                    # dist[i] = P(sum of dice so far == i)
for _ in range(n):
    newdist = [0.0] * (len(dist) + 6)
    for face in range(1, 7):
        for i, p in enumerate(dist):
            newdist[i + face] += p / 6
    dist = newdist
exact  = sum(dist[k:])          # dist[i] indexes the sum value directly
markov = mu / k
cheby  = var / (k - mu) ** 2
print(f"exact={exact:.4f}  Markov<={markov:.4f}  Chebyshev<={cheby:.4f}")
```

Output. `exact=0.1075  Markov<=0.8750  Chebyshev<=0.5833` — both are valid upper bounds on the true value, and Chebyshev is somewhat tighter here since it also exploits the variance of S.

Convex and concave functions

Definition 2.2.49 (Convex function)

$f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ is *convex* if for all $x_1, x_2 \in \mathbb{R}^n$ and $\theta \in [0, 1]$,

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta) f(x_2)$$

(*strictly* convex if the inequality is strict.)

Plain English. For $f : \mathbb{R} \to \mathbb{R}$: given any two points $P = (x_1, f(x_1))$ and $Q = (x_2, f(x_2))$ on the graph of f, the straight line segment from P to Q always sits *above* (or on) the graph of f. Examples of convex functions: x^2, e^x, $-\log(x)$, $-\sqrt{x}$.

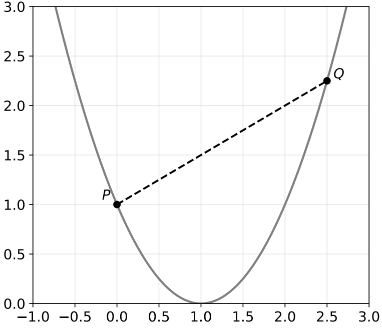
Definition 2.2.50 (Concave function)

f is *concave* if the inequality reverses: $f(\theta x_1 + (1-\theta)x_2) \geq \theta f(x_1) + (1-\theta) f(x_2)$. Note f is (strictly) concave iff $-f$ is (strictly) convex.

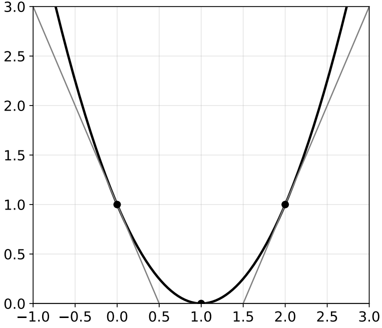
Lemma 2.2.51, illustrated: convexity via secants and tangents

Lemma 2.2.51 (First-order convexity conditions)

For convex $f : \mathbb{R}^n \to \mathbb{R}$, for all $x, x_0 \in \mathbb{R}^n$, $f(x) \geq f(x_0) + (x - x_0) \cdot \nabla f(x_0)$, where ∇ (nabla) denotes a sub-gradient of f at x_0: the tangent line at any point lies *below* the function everywhere else.



Secant PQ sits *above* the curve.



Tangents at $x_0 = 0, 1, 2$ sit *below* the curve.

Theorem 2.2.52: Jensen's inequality

Theorem 2.2.52 (Jensen's inequality)

Let X be a random vector. Then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] \quad \text{for convex } f \qquad\qquad g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)] \quad \text{for concave } g$$

Plain English. Applying a convex function *after* averaging never overshoots applying the function *first* and then averaging: the function of the average is at most the average of the function. (For concave functions, the inequality flips.)

Proof (convex case)

By Lemma 2.2.51 with $x_0 = \mathbb{E}[X]$: $f(x) \geq f(\mathbb{E}[X]) + (x - \mathbb{E}[X]) \cdot \nabla f(\mathbb{E}[X])$. Take expectations of both sides:

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) + \nabla f(\mathbb{E}[X]) \cdot \underbrace{\mathbb{E}[X - \mathbb{E}[X]]}_{=0} = f(\mathbb{E}[X])$$

(The concave case follows by applying the convex result to $-g$.)

Corollary 2.2.54: Jensen's inequality of averages

Corollary 2.2.54 (Jensen's inequality of averages)

Let f be convex, and $(x_i)_{i=1}^n$ a finite sequence. Then

$$f\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) \leq \frac{1}{n}\sum_{i=1}^{n} f(x_i)$$

Plain English. This is Jensen's inequality specialized to a uniform average over n numbers: applying convex f to the plain average of the x_i never exceeds the plain average of $f(x_i)$.

Proof

Let $p = (1/n, \ldots, 1/n)$ and let X have sample space $\{x_1, \ldots, x_n\}$ with $\mathrm{P}(X = x_i) = 1/n$. Then $\mathbb{E}[X] = \frac{1}{n}\sum_i x_i$ and $\mathbb{E}[f(X)] = \frac{1}{n}\sum_i f(x_i)$, so Theorem 2.2.52 (Jensen's inequality) directly gives

$$f\left(\frac{1}{n}\sum_i x_i\right) = f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] = \frac{1}{n}\sum_i f(x_i)$$

Motivation: what does it mean for random variables to converge?

For an ordinary sequence of numbers (x_n), "convergence" is unambiguous. For a sequence of *functions* (f_n), there are already several distinct notions: pointwise convergence ($\forall x, \lim_n f_n(x) = f(x)$), uniform convergence ($\lim_n \sup_x |f_n(x) - f(x)| = 0$), and L^2 convergence ($\lim_n \int |f_n(x) - f(x)|^2 dx = 0$). Since random variables are themselves functions $\Omega \to \mathbb{R}$, we might try to directly carry across the notion of pointwise convergence:

Definition 2.2.55 (Pointwise convergence of random variables)

$(X_n)_{n=1}^{\infty}$ converges *pointwise* to X if for all $\omega \in \Omega$, $\lim_{n\to\infty} X_n(\omega) = X(\omega)$.

Plain English. This turns out to be too strong: for a "poor" choice of the sampled outcome ω, the sequence $\{X_n(\omega)\}_{n=1}^{\infty}$ may converge to something else entirely, or not converge at all — even though such ω may themselves be extremely unlikely to ever be sampled.

Example 2.2.56: pointwise (non)convergence for coin flips

Example 2.2.56 (Bernoulli pointwise (non)convergence)

Let X_n be i.i.d. with $\{0, 1\}$ and probabilities $\{1 - \theta, \theta\}$ (modelling the n-th flip of a biased coin). Define $S_n(\omega_{1:\infty}) = \sum_{i=1}^{n} X_i(\omega_i)$ (number of 1's in the first n tosses). We would hope

$$\forall \omega \in \Omega^\infty, \quad \lim_{n \to \infty} \frac{S_n(\omega)}{n} = \theta$$

Plain English. This says: in the long run, the fraction of heads should settle at θ. But this fails for *every* choice of ω: e.g. for $\omega_z = 000\ldots$ (the all-zeros sequence), $\lim_{n\to\infty} S_n(\omega_z)/n = 0 \neq \theta$. Yet we would not expect a fair coin to actually produce an all-zeros sequence — this particular ω_z is simply "unlikely". This motivates weakening pointwise convergence to allow a probability-zero set of exceptions.

Definition 2.2.57: almost sure convergence

Definition 2.2.57 (Almost sure convergence (a.s.))

$(X_n)_{n=1}^{\infty}$ converges *almost surely* to X (denoted $X_n \xrightarrow{\text{a.s.}} X$) if

$$\mathrm{P}\left[\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}\right] = 1$$

Plain English. Rather than requiring convergence for *every* ω, we only require convergence for a set of outcomes with total probability 1 — i.e. convergence happens "almost everywhere", ignoring a probability-zero set of unlucky exceptions. Also called *strong convergence* or *convergence with probability 1 (w.p.1)*. The set of all counterexamples $\{\omega : X_n(\omega) \not\to X(\omega)\}$ is a probability-zero event, so even though counterexamples may exist, we will *almost surely* never see one.

Example 2.2.58

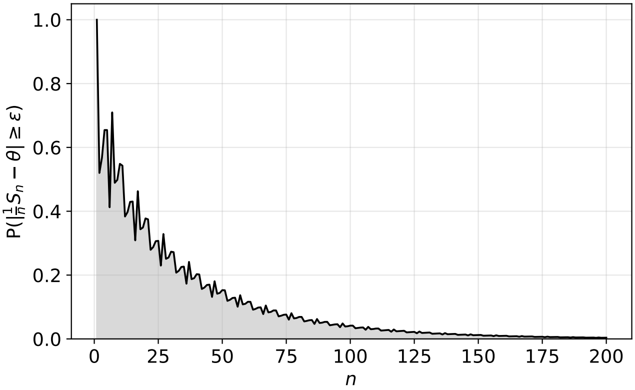
For a biased coin ($0 < \theta < 1$), define $X_n(\omega_{1:\infty}) = [\![\exists i \leq n,\, \omega_i = 1]\!]$ ("at least one head seen in the first n flips"). Writing the event $\{\omega : \exists i, \omega_i = 1\}$ as the disjoint union $\bigcup_{i=0}^{\infty} \Gamma_{0^i 1}$ of cylinder sets ("i zeros then a one"),

$$\mathrm{P}\left(\{\omega : \lim_{n\to\infty} X_n(\omega) = 1\}\right) = \sum_{i=0}^{\infty} \mathrm{P}(\Gamma_{0^i 1}) = \sum_{i=0}^{\infty} (1-\theta)^i \theta = \frac{\theta}{1-(1-\theta)} = 1$$

Plain English. Since $\theta > 0$, the probability of an infinite run of all-zeros (never flipping a single head) is zero, so with probability 1 the coin eventually flips at least one head — hence $X_n \xrightarrow{\text{a.s.}} 1$.

Remark 2.2.59 (Weak law of large numbers, informally)

For i.i.d. (X_i) with $X = 1$ w.p. θ and 0 w.p. $1 - \theta$ (a *Bernoulli process*), $\mathbb{E}[X] = \theta$ and $\mathrm{Var}[X] = \theta(1 - \theta)$. Let $S_n = \sum_{i=1}^{n} X_i$. We can now repair Example 2.2.56: the set of counterexample ω for which $\frac{1}{n} S_n(\omega) \not\to \theta$ forms a probability-zero event.



Theorem 2.2.60: the Weak Law of Large Numbers

Theorem 2.2.60 (Weak law of large numbers)

Let S_n be a sum of n i.i.d. Bernoulli(θ) random variables. Then for all $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathrm{P}\left(\left|\frac{S_n}{n} - \theta\right| \geq \varepsilon\right) = 0$$

Plain English. The probability distribution of the sample average S_n/n concentrates *all* of its probability mass onto the single point θ as $n \to \infty$.

Proof

Let $Y = \frac{1}{n} S_n$. By linearity, $\mathbb{E}[Y] = \theta$, and by Theorem 2.2.43, $\mathrm{Var}[Y] = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}[X_i] = \frac{\theta(1-\theta)}{n}$. By Chebyshev's inequality, and using $\theta(1-\theta) \leq \frac{1}{4}$ for $\theta \in [0,1]$,

$$\mathrm{P}\left(\left|\frac{S_n}{n} - \theta\right| \geq \varepsilon\right) = \mathrm{P}(|Y - \mathbb{E}[Y]| \geq \varepsilon) \leq \frac{\mathrm{Var}[Y]}{\varepsilon^2} = \frac{\theta(1-\theta)}{\varepsilon^2 n} \leq \frac{1}{4\varepsilon^2 n}$$

which tends to 0 as $n \to \infty$.

Theorem 2.2.61 (Hoeffding bound)

Let S_n be a sum of n independent $[0,1]$-valued random variables. Then

$$\mathrm{P}\big(|S_n - \mathbb{E}[S_n]| \geq t\big) \leq 2e^{-2t^2/n}$$

Plain English. The proof of the weak law gave a convergence rate of $O(n^{-1})$ (Chebyshev). Hoeffding's bound gives an *exponentially* better rate: setting $t = n\varepsilon$ for i.i.d. $X_i \in [0,1]$ with $\mathbb{E}[X_i] = \theta$,

$$\mathrm{P}\left(\left|\frac{1}{n}S_n - \theta\right| \geq \varepsilon\right) \leq 2e^{-2n\varepsilon^2}$$

which decays exponentially fast in n, rather than only like $1/n$.

Limit supremum and limit infimum of events

Definition 2.2.62 (Limit supremum / Limit infimum)

For a sequence of events $(A_n)_{n=1}^{\infty}$,

$$\limsup_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} \bigcup_{j=n}^{\infty} A_j \qquad \liminf_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} \bigcap_{j=n}^{\infty} A_j$$

Remark 2.2.63 (Infinitely often and almost all)

$\omega \in \limsup A_n$ iff $\omega \in A_j$ for infinitely many j ("A_n infinitely often"). $\omega \in \liminf A_n$ iff $\omega \in A_j$ for all j beyond some point n ("A_n almost all", i.e. all but finitely many events occur).

Plain English. $\limsup A_n$ collects outcomes that keep re-appearing in the sequence of events *forever* (infinitely often, even if not every single time); $\liminf A_n$ collects outcomes for which *eventually every subsequent event occurs* (almost all of the time, past some point). Example 2.2.64: with the biased-coin sample space, one can show the coin will almost surely flip a 1 infinitely often.

Convergence in probability

Definition 2.2.65 (Convergence in probability (i.p.))

$(X_n)_{n=1}^{\infty}$ converges *in probability* (also *weak convergence*) to X, denoted $X_n \xrightarrow{P} X$, if for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathrm{P}(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) = 0$$

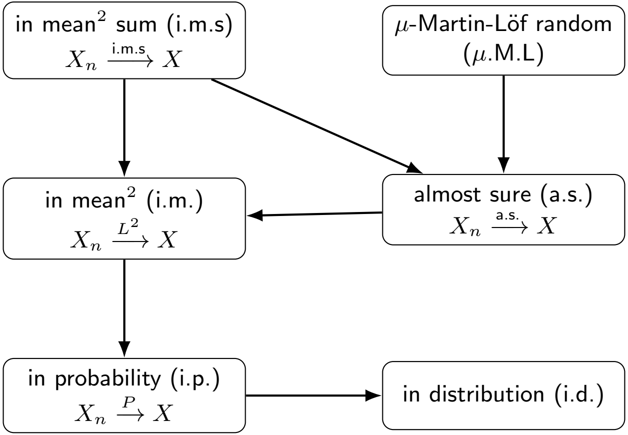
Plain English. Weaker than almost sure convergence: it only requires the *probability* of a large deviation to vanish at each fixed n (not that the sample paths themselves settle down). Theorem 2.2.60 shows $\frac{1}{n}S_n \xrightarrow{P} \theta$ — the weak law of large numbers is, formally, an instance of convergence in probability.

Theorem 2.2.66 (Convergence a.s. implies convergence i.p.)

If $X_n \xrightarrow{\text{a.s.}} X$, then $X_n \xrightarrow{P} X$.

Plain English. Almost sure convergence is *strictly stronger* than convergence in probability: it implies i.p. convergence, but — as Example 2.2.67 shows next — the converse fails. This is why almost sure convergence is called "strong" and convergence in probability is called "weak".

The convergence hierarchy (Figure 2.8)



Plain English. An arrow $A \to B$ means "A-convergence implies B-convergence". Convergence in mean^2 sum is the strongest notion shown; convergence in distribution is the weakest. Notably, almost sure convergence and convergence in mean^2 each imply convergence in probability, but neither is implied by it (as the next example shows).

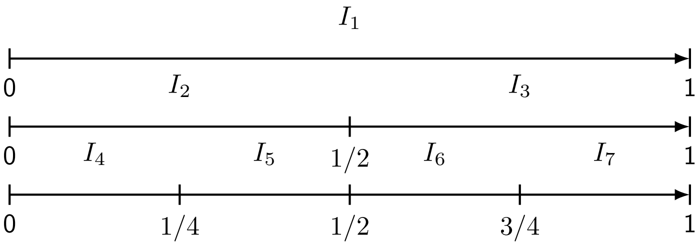
Example 2.2.67

On $\Omega = [0,1]$ (Example 2.2.25), define binary intervals

$$I_{2^m+i} = \left[\frac{i}{2^m}, \frac{i+1}{2^m}\right), \qquad m = 0,1,2,\ldots, \; i = 0,\ldots,2^m-1$$

For each m there are 2^m intervals of length 2^{-m} covering $[0,1]$. As $m \to \infty$, $\mathrm{P}(I_n) = \mathrm{P}(I_{2^m}) = 2^{-m} \to 0$. Define $Y_n(\omega) = [\![\omega \in I_n]\!]$.

Example 2.2.67: convergence i.p. does not imply convergence a.s. II



Plain English (figure, Figure 2.9 in book). At level m, the unit interval is chopped into 2^m equal binary sub-intervals $I_{2^m}, \ldots, I_{2^{m+1}-1}$; as m grows, each individual sub-interval shrinks to length 2^{-m}, so its probability under the uniform measure shrinks to 0.

Example 2.2.67: convergence i.p. does not imply convergence a.s. III

For any $0 < \varepsilon \leq 1$,

$$\lim_{n\to\infty} \mathrm{P}(|Y_n - 0| > \varepsilon) = \lim_{n\to\infty} \mathrm{P}(Y_n = 1) = \lim_{n\to\infty} \mathrm{P}(I_n) = 0$$

so $Y_n \xrightarrow{P} 0$. **However**, $Y_n \overset{\text{a.s.}}{\not\longrightarrow} 0$: for *every* $\omega \in [0,1)$, since $I_{2^m}, \ldots, I_{2^{m+1}-1}$ partitions $[0,1)$ at every level m, ω belongs to exactly one interval at each level, so $Y_n(\omega) = 1$ infinitely often (once per level m), and the limit $\lim_{n\to\infty} Y_n(\omega)$ never settles — it diverges for *every* ω:

$$\mathrm{P}\big(\{\omega \in \Omega : \lim_{n\to\infty} Y_n(\omega) = 0\}\big) = \mathrm{P}(\emptyset) = 0 \neq 1$$

Plain English. This is the promised counterexample: convergence in probability holds (the probability of a deviation shrinks to 0 with n), yet almost sure convergence completely fails (in fact, *every* sample path fails to converge) — confirming that i.p. is strictly weaker than a.s.

Convergence in mean^2 and in mean^2 sum I

Definition 2.2.69 (Convergence in mean^2 (i.m.))

(X_n) converges *in mean*^{2} to X (denoted $X_n \xrightarrow{L^2} X$) if $\lim_{n\to\infty} \mathbb{E}[(X_n - X)^2] = 0$. ("Mean^2" is pronounced "in mean square"; not a footnote superscript.)

Proposition 2.2.70 (Convergence in mean^2 implies convergence in probability)

If $X_n \xrightarrow{L^2} X$, then $X_n \xrightarrow{P} X$.

Proof. By Markov's inequality applied to $|X_n - X|^2$:

$$\mathrm{P}(|X_n - X| > \varepsilon) = \mathrm{P}(|X_n - X|^2 > \varepsilon^2) \leq \frac{1}{\varepsilon^2}\mathbb{E}[|X_n - X|^2] \longrightarrow 0$$

Convergence in mean^2 and in mean^2 sum II

Example 2.2.71 (i.p. does not imply i.m.)

Let $X_n = n$ w.p. $1/n$, else 0. Then $\mathrm{P}(|X_n| > \varepsilon) = 1/n \to 0$ so $X_n \xrightarrow{P} 0$, but $\mathbb{E}[X_n^2] = n^2 \cdot \frac{1}{n} = n \to \infty$, so $X_n \not\xrightarrow{L^2} 0$.

Convergence in mean^2 and in mean^2 sum III

Definition 2.2.72 (Convergence in mean^2 sum (i.m.s))

$(X_n)_{n=1}^{\infty}$ (non-negative) converges *in mean^2 sum* to X if $\sum_{n=1}^{\infty} \mathbb{E}[(X_n - X)^2] < \infty$.

Proposition 2.2.73 (Conv. in mean^2 sum implies conv. in mean^2)

Trivial: an infinite sum of non-negative terms can only be finite if the summands themselves tend to zero.

Plain English. Convergence in mean^2 sum is a strictly stronger notion: not only must $\mathbb{E}[(X_n - X)^2] \to 0$, but these quantities must shrink *fast enough* that their infinite sum stays finite.

Borel-Cantelli Lemma, and convergence i.m.s. implies a.s.

Lemma 2.2.74 (Borel-Cantelli)

If $\sum_{n=1}^{\infty} \mathrm{P}(E_n) < \infty$, then $\mathrm{P}(\limsup_{n\to\infty} E_n) = 0$: the probability that infinitely many of $E_1, E_2, \ldots$ occur is zero.

Plain English. If the probabilities of a sequence of events shrink fast enough to sum to something finite, then (almost surely) only *finitely many* of those events can ever actually happen.

Theorem 2.2.75 (Convergence i.m.s. implies convergence a.s.)

If $X_n \xrightarrow{\text{i.m.s}} X$, then $X_n \xrightarrow{\text{a.s.}} X$.

Proof sketch. Let $Y_n = X_n - X$ and $E_n = \{\omega : Y_n(\omega)^2 \geq \varepsilon^2\}$. By Markov's inequality, $\mathrm{P}(E_n) \leq \mathbb{E}[Y_n^2]/\varepsilon^2$, so $\sum_n \mathrm{P}(E_n) \leq \frac{1}{\varepsilon^2} \sum_n \mathbb{E}[Y_n^2] < \infty$ by the i.m.s. assumption. Borel-Cantelli then gives $\mathrm{P}(\limsup_n E_n) = 0$, which (unpacking Definition 2.2.57(ii)) is exactly almost sure convergence $Y_n \xrightarrow{\text{a.s.}} 0$, i.e. $X_n \xrightarrow{\text{a.s.}} X$.

Python: simulating the weak law of large numbers and Hoeffding's bound

We simulate a biased coin and empirically check the gap between Chebyshev's ($O(1/n)$) and Hoeffding's (exponential) tail bounds.

```
import random, math
theta, eps, trials = 0.4, 0.1, 200_000
def deviation_freq(n):
    """Empirical P(|S_n/n - theta| >= eps) over `trials` runs."""
    bad = 0
    for _ in range(trials):
        S = sum(1 for _ in range(n) if random.random() < theta)
        if abs(S / n - theta) >= eps:
            bad += 1
    return bad / trials
for n in [50, 150, 300]:
    empirical = deviation_freq(n)
    chebyshev = min(1.0, theta * (1 - theta) / (eps ** 2 * n))
    hoeffding = min(1.0, 2 * math.exp(-2 * n * eps ** 2))
    print(f"n={n:4d}  empirical={empirical:.4f}"
          f"  Chebyshev<={chebyshev:.4f}  Hoeffding<={hoeffding:.4f}")
```

Typical output. `n=50  empirical=0.1533  Chebyshev<=0.4800  Hoeffding<=0.7358`; `n=300 empirical=0.0003  Chebyshev<=0.0800  Hoeffding<=0.0050` — both bounds stay valid, and Hoeffding decays exponentially faster, as Theorem 2.2.61 predicts.

Chapter 2.2 summary

- **Foundations (2.2.1–2.2.2).** A σ-algebra $\mathcal{F}$ picks out which subsets of Ω count as measurable "events"; a probability (semi)measure P assigns them numbers in $[0, 1]$ satisfying countable additivity. Cylinder sets Γ_x and (semi)measures ν on $\mathcal{X}^\infty$ are the central objects used throughout the rest of the book.
- **Conditioning and random variables (2.2.3–2.2.5).** Conditional probability $\mathrm{P}(B|A)$ restricts attention to A; a random variable $X : \Omega \to \mathbb{R}$ turns outcomes into numbers, inducing a cdf/pmf/pdf; joint and conditional distributions extend this to several variables at once, governed by the sum and product rules.
- **Expectation, variance, inequalities (2.2.6–2.2.7).** $\mathbb{E}[X]$ is the probability-weighted average; $\mathrm{Var}[X]$ measures spread around it; Markov, Chebyshev, and Jensen's inequalities bound tail probabilities and convex transformations using only these summary statistics.
- **Convergence (2.2.8).** Random variables can converge to a limit in several inequivalent senses — almost surely, in probability, in mean^2, in mean^2 sum, in distribution — forming a strict hierarchy (Figure 2.8). The weak law of large numbers and the exponentially sharper Hoeffding bound both formalize "sample averages concentrate around the true mean" as $n \to \infty$.